```
import pandas as pd
import numpy as np

from sklearn.model_selection import train_test_split
from sklearn.preprocessing import LabelEncoder
from sklearn.metrics import mean_absolute_error, mean_squared_error, r2_score

from xgboost import XGBRegressor
import matplotlib.pyplot as plt
```

```
df = pd.read_csv("car data.csv")
print(df.head())
```

```
  Car_Name  Year  Selling_Price  Present_Price  Kms_Driven Fuel_Type  \
0     ritz  2014           3.35           5.59       27000    Petrol
1      sx4  2013           4.75           9.54       43000    Diesel
2     ciaz  2017           7.25           9.85        6900    Petrol
3  wagon r  2011           2.85           4.15        5200    Petrol
4    swift  2014           4.60           6.87       42450    Diesel

  Seller_Type Transmission  Owner
0      Dealer       Manual      0
1      Dealer       Manual      0
2      Dealer       Manual      0
3      Dealer       Manual      0
4      Dealer       Manual      0
```

```
df.drop("Car_Name", axis=1, inplace=True)
```

```
df["Car_Age"] = 2026 - df["Year"]
df.drop("Year", axis=1, inplace=True)
```

```
le = LabelEncoder()

for col in ["Fuel_Type", "Seller_Type", "Transmission"]:
    df[col] = le.fit_transform(df[col])
```

```
X = df.drop("Selling_Price", axis=1)
y = df["Selling_Price"]
```

```
X_train, X_test, y_train, y_test = train_test_split(
    X, y, test_size=0.2, random_state=42
)
```

```
model = XGBRegressor(
    n_estimators=200,
    learning_rate=0.05,
    max_depth=4,
    random_state=42
)

model.fit(X_train, y_train)
```

```
XGBRegressor

XGBRegressor(base_score=None, booster=None, callbacks=None,
             colsample_bylevel=None, colsample_bynode=None,
             colsample_bytree=None, device=None, early_stopping_rounds=None,
             enable_categorical=False, eval_metric=None, feature_types=None,
             feature_weights=None, gamma=None, grow_policy=None,
             importance_type=None, interaction_constraints=None,
             learning_rate=0.05, max_bin=None, max_cat_threshold=None,
             max_cat_to_onehot=None, max_delta_step=None, max_depth=4,
             max_leaves=None, min_child_weight=None, missing=nan,
             monotone_constraints=None, multi_strategy=None, n_estimators=200,
             n_jobs=None, num_parallel_tree=None, ...)
```

```
y_pred = model.predict(X_test)
```

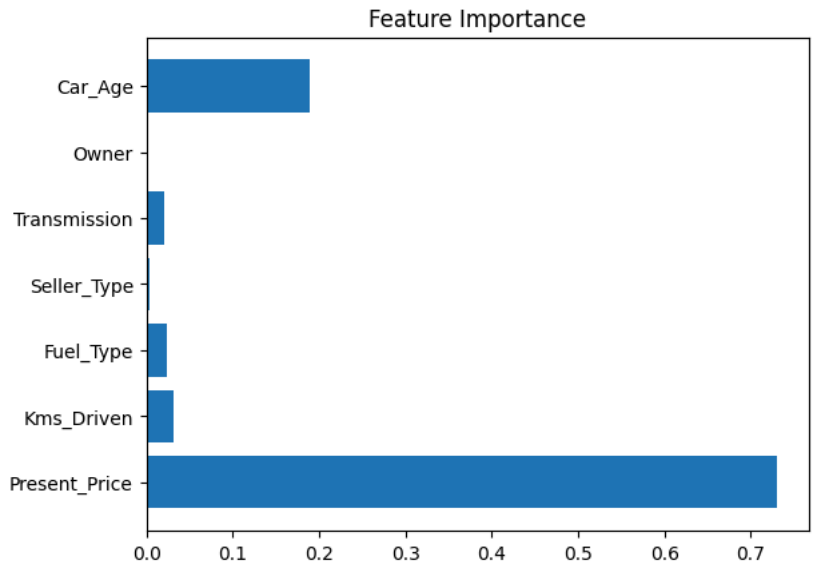
```
mae = mean_absolute_error(y_test, y_pred)
mse = mean_squared_error(y_test, y_pred)
rmse = np.sqrt(mse)
r2 = r2_score(y_test, y_pred)
```

```
print("MAE:", mae)
print("MSE:", mse)
print("RMSE:", rmse)
print("R2 Score:", r2)
```

```
MAE: 0.5976336962864047
MSE: 1.0332376785228286
RMSE: 1.016482994704205
R2 Score: 0.9551460186472486
```

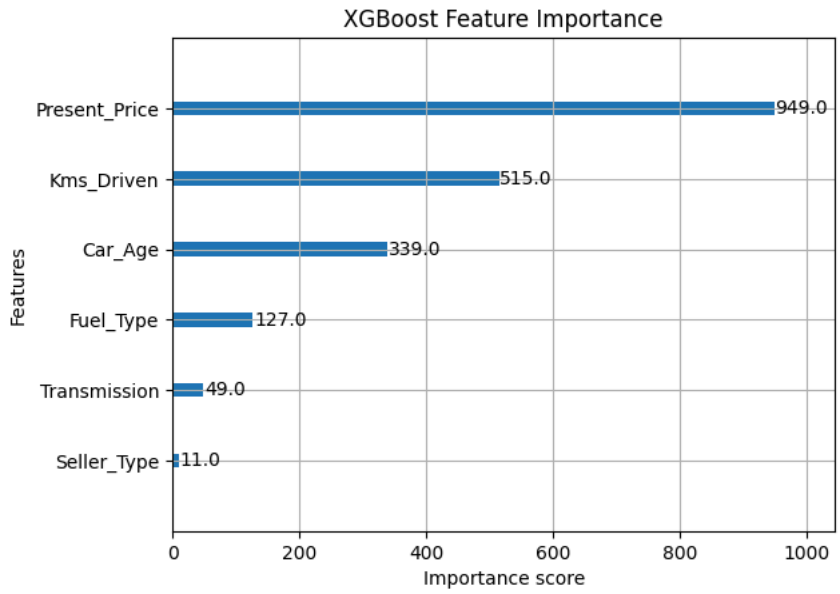
```
importance = model.feature_importances_

plt.barh(X.columns, importance)
plt.title("Feature Importance")
plt.show()
```




```
from xgboost import plot_importance
import matplotlib.pyplot as plt

plot_importance(model)
plt.title("XGBoost Feature Importance")
plt.show()
```



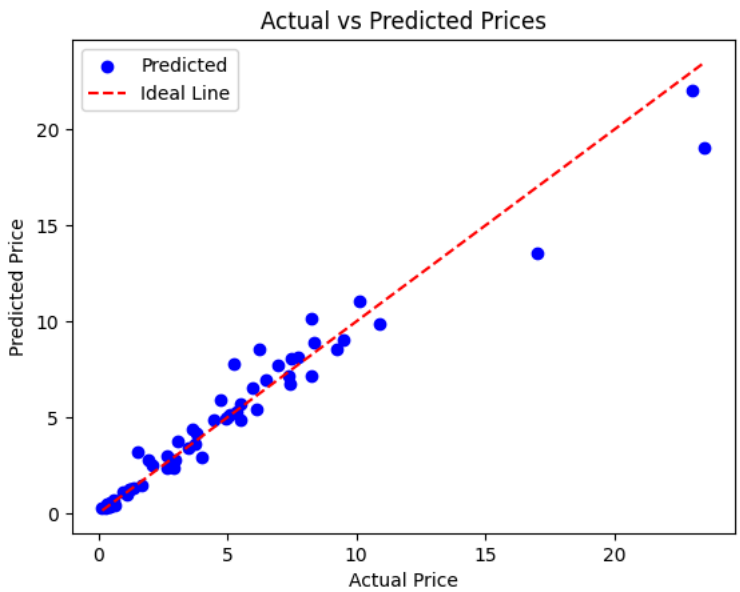

```
from xgboost import plot_tree
import matplotlib.pyplot as plt

plt.figure(figsize=(20,10))
plot_tree(model, num_trees=0)
plt.show()
```

```
/usr/local/lib/python3.12/dist-packages/xgboost/plotting.py:268: FutureWarning: The `num_trees` parameter is deprecated, use `tree_idx` insetad
  warnings.warn(
<Figure size 2000x1000 with 0 Axes>
```

```
plt.scatter(y_test, y_pred, color='blue', label='Predicted')
plt.plot([y_test.min(), y_test.max()],
         [y_test.min(), y_test.max()],
         color='red', linestyle='--', label='Ideal Line')

plt.xlabel("Actual Price")
plt.ylabel("Predicted Price")
plt.title("Actual vs Predicted Prices")
plt.legend()
plt.show()
```



Start coding or generate with AI.